

## Lec 28

- ① One function of two R.V.s.
- ② Two functions of two R.V.s

→ Examples:

①  $Z = X + Y$

②  $Z = X/Y$

Give joint density of

$X, Y$   $f_{X,Y}(x,y)$

- find  $f_Z(z)$

$$② Z = X + Y$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z-y, y) dy$$

If  $X$  and  $Y$  are independent

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy$$

$X, Y$  are iid  $\exp(\lambda)$

$$f_X(x) = \lambda e^{-\lambda x} \quad x \geq 0$$

$$f_{X,Y}(x,y) = \lambda e^{-\lambda x} \lambda e^{-\lambda y} \quad x \geq 0, y \geq 0$$

$$* X \sim \text{Gamma}(\lambda, \alpha)$$

$$\text{where } \Gamma(\alpha) = \int_0^{\infty} \lambda^{\alpha} x^{\alpha-1} e^{-\lambda x} dx$$

$$f_Z(z) = \int_0^z \lambda^2 e^{-\lambda(z-y)} e^{-\lambda y} dy$$

$$= \int_0^z \lambda^2 e^{-\lambda z} dy$$

$$\begin{matrix} z-y \geq 0 \\ y \geq 0 \end{matrix} \Rightarrow \lambda^2 e^{-\lambda z} \int_0^z dy$$

$$\boxed{z \geq y \geq 0} \Rightarrow \lambda^2 z e^{-\lambda z}$$

$$Z \sim \text{Gamma}(\lambda, 2)$$

$$W \sim \text{Gamma}(\lambda, n) \quad f_W(w) = \frac{\lambda^n}{\Gamma(n)} w^{n-1} e^{-\lambda w} \quad w \geq 0$$

$$f_X(x) = \frac{\lambda^{\alpha} x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$$

When  $\lambda$  is an integer it is referred to as an Erlang distribution.

Suppose  $Y \sim \text{Gamma}(\lambda, n-1)$   
 $X \sim \text{Exponential}(\lambda)$

$$Z = X + Y$$

$$\begin{aligned} f_Z(z) &= \int_0^z f_X(z-y) f_Y(y) dy \\ &= \int_0^z \lambda e^{-\lambda(z-y)} \frac{\lambda^{n-1} y^{n-2} e^{-\lambda y}}{(n-2)!} dy \\ &= \frac{\lambda^n e^{-\lambda z}}{(n-2)!} \int_0^z y^{n-2} dy \\ &= \frac{\lambda^n e^{-\lambda z}}{(n-2)!} \left( \frac{y^{n-1}}{n-1} \right) \bigg|_0^z = \frac{\lambda^{n-1} z^{n-2} e^{-\lambda z}}{(n-1)!} \end{aligned}$$

$$Z \sim \text{Gamma}(\lambda, n)$$

$$Z = X_1 + X_2 + \dots + X_n$$

where  $X_i$ 's are iid  $\text{Exp}(\lambda)$  random variables

$$X \sim N(\mu_1, \sigma_1^2) \quad Y \sim N(\mu_2, \sigma_2^2)$$

Suppose  $X, Y$  are independent

find pdf of  $Z = X + Y$

$$Z \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$X, Y \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, r)$$

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = r$$

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$$X \sim N(0, 1), Y \sim N(0, 1)$$

$X, Y$  are independent

$$f_{X,Y}(x,y) = \frac{e^{-\frac{(x^2+y^2)}{2}}}{2\pi} \quad x, y \in \mathbb{R}$$

$$Z = X + Y \quad f_X(x)f_Y(y)$$

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy \\ &= \int_{-\infty}^{\infty} \frac{e^{-\left(\frac{(z-y)^2 + y^2}{2}\right)}}{2\pi} dy \end{aligned}$$

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{z^2 + 2yz + y^2}{2}\right)} dy$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \frac{1}{\sqrt{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-z}{\sqrt{2}}\right)^2} dy$$

$$= \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi} \sqrt{2}}$$

$$Z \sim N(0, (\sqrt{2})^2)$$

$$\sigma_z = \sqrt{2}$$

$$-\frac{1}{2} \left( \frac{y^2 - yz + \frac{z^2}{4}}{\frac{1}{2}} \right) = -\frac{z^2}{4}$$

$$-\frac{1}{2} \left[ \frac{zy^2 - 2yz + \frac{z^2}{2}}{\frac{1}{2}} \right] = -\frac{z^2}{4}$$

$$X \sim \text{Binomial}(n, p)$$

$$X+Y \sim \text{Binomial}(n+m, p)$$

$$Y \sim \text{Binomial}(m, p)$$

Example

$$R = \dots$$

$$\theta = \dots$$



dy

Two functions of two R.v.s

$$Z = g_1(x, y)$$

$$W = g_2(x, y)$$

To find  $f_{Z,W}(z, w)$  given  $f_{X,Y}(x, y)$

$$f_{Z,W}(z, w) = \frac{f_{X,Y}(x, y)}{J_{x,y}(z, w)} \quad \begin{matrix} x, y \text{ st} \\ z = g_1(x, y) \\ w = g_2(x, y) \end{matrix}$$

$$J_{x,y}(z, w) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix}$$

Example:

$$R = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\Leftrightarrow \begin{cases} x = R \cos \theta \\ y = R \sin \theta \end{cases}$$

$$R = \sqrt{x^2 + y^2}$$

$$J_{x,y}(r, \theta) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$Y = g(X)$$

$$f_Y(y) = \frac{f_X(x)}{g'(x)} \quad x: g(x)=y$$

$$= f_X(x) |J(x)|$$

## ② Linear transformation.

$$Z = aX + bY$$

$$W = cX + dY$$

Find  $f_{Z,W}(z,w)$  from  $f_{X,Y}(x,y)$ .

$$\begin{bmatrix} Z \\ W \end{bmatrix} = \underbrace{\begin{bmatrix} a & b \\ c & d \end{bmatrix}}_A \begin{bmatrix} X \\ Y \end{bmatrix}$$

Assume  $A$  is invertible.

Thus  $\det(A) \neq 0$ .

$$J(z,w) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \det(A)$$

$$f_{Z,W}(z,w) = \frac{f_{X,Y}(x,y)}{\det(A)}$$

### Exercises:

①  $X, Y \sim N(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$

Show that  $Z = X + Y$  is a normal distribution.

$$Z \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y)$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y)$$

②  $(X, Y) \sim N(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$

$$\begin{bmatrix} Z \\ W \end{bmatrix} = \underbrace{\begin{bmatrix} a & b \\ c & d \end{bmatrix}}_A \begin{bmatrix} X \\ Y \end{bmatrix}, \quad A \text{ is invertible?}$$

Show that  $Z, W$  are jointly Gaussian.

$$(Z, W) \sim N\left(\begin{bmatrix} a\mu_X + b\mu_Y \\ c\mu_X + d\mu_Y \end{bmatrix}, \begin{bmatrix} a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y & a c\sigma_X^2 + b d\sigma_Y^2 + (a d + b c)\rho\sigma_X\sigma_Y \\ a c\sigma_X^2 + b d\sigma_Y^2 + (a d + b c)\rho\sigma_X\sigma_Y & c^2\sigma_X^2 + d^2\sigma_Y^2 + 2cd\rho\sigma_X\sigma_Y \end{bmatrix}\right)$$

$$E[Z] = E[aX + bY] = a\mu_X + b\mu_Y$$

$$\text{Var}(Z) = \text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X,Y)$$

$$\text{Cov}(Z, W) = E[(Z - E[Z])(W - E[W])] = a c\sigma_X^2 + b d\sigma_Y^2 + (a d + b c)\rho\sigma_X\sigma_Y$$

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Example 3:

$$Z = \max\{X, Y\}, \quad W = \min\{X, Y\}$$

Suppose  
(a)  $X, Y$  are independent

$$F_Z(z) = P(Z \leq z)$$

$$= P(X \leq z, Y \leq z)$$

$$= F_X(z) F_Y(z)$$

$$f_Z(z) = \frac{\partial F_Z(z)}{\partial z} = f_X(z) F_Y(z) + F_X(z) f_Y(z)$$



$$\begin{aligned}
 1 - F_W(w) &= P(W > w) \\
 &= P(X > w, Y > w) \quad \uparrow X, Y \text{ are independent} \\
 &= P(X > w) P(Y > w) \\
 &= (1 - F_X(w)) (1 - F_Y(w))
 \end{aligned}$$

$$-f_W(w) = -f_X(w) (1 - F_Y(w)) - f_Y(w) (1 - F_X(w))$$

$$\Rightarrow f_W(w) = f_X(w) (1 - F_Y(w)) + f_Y(w) (1 - F_X(w))$$

(a.1) Suppose  $X \sim \exp(\lambda_1)$ ,  $Y \sim \exp(\lambda_2)$   
 $\min(X, Y) \sim \exp(\lambda_1 + \lambda_2)$  (Exercise)